

KHOUN KHAN, Thailand

WMO: 485700

Lat: 6.6536N

Lon: 100.0831E

Elev: 20

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	73.0	74.0	65.0	92.7	84.7	66.5	97.6	84.1	10.6	85.4	9.1	86.4	0.1	N/A	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	17.2	95.8	77.8	94.4	78.0	93.2	78.2	81.4	90.2	80.9	89.4	80.4	88.7	3.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	151.4	85.2	78.7	149.0	84.8	78.2	146.6	84.3	45.1	90.1	44.5	89.6	44.0	89.0	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	8.5	7.4	DB	69.9	97.5	1.8	1.8	68.6	98.8	67.5	99.9	66.5	101.0	65.1	102.3	
			WB	67.0	82.9	2.0	0.8	65.6	83.5	64.4	84.0	63.2	84.4	61.8	85.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	82.9	84.0	84.8	84.9	84.1	83.3	82.6	82.4	82.1	82.0	82.4	82.4
	DBStd	2.04	1.82	1.73	1.96	1.75	1.66	1.63	1.64	1.84	1.95	1.73	1.80	1.80
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12103	1021	953	1078	1048	1056	1000	1010	1006	962	993	971	1005
	CDD65	6628	556	533	613	598	591	550	545	541	512	528	521	540
	CDH74	70765	5925	6030	7106	6847	6668	6086	5768	5597	5042	5024	5131	5541
	CDH80	27700	2295	2732	3376	3060	2706	2397	2152	2023	1703	1608	1734	1914
Wind	WSAvg	1.3	2.5	2.5	1.4	0.8	0.7	1.0	1.1	1.2	1.0	0.9	1.0	1.9
Precipitation	PrecAvg	80.6	2.0	1.8	5.2	6.7	7.1	6.5	7.5	8.9	10.2	11.3	8.4	5.2
	PrecMax	95.0	11.6	6.6	15.9	12.0	13.1	13.4	13.1	13.3	17.0	20.1	18.5	19.2
	PrecMin	61.0	0.3	0.0	0.2	0.9	3.4	1.1	3.0	1.6	5.9	5.2	4.1	0.0
	PrecStd	8.0	2.4	1.8	3.7	2.8	2.2	2.6	2.6	3.0	2.8	3.1	3.5	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	96.0	98.1	97.7	95.3	93.2	92.2	92.2	92.2	92.0	92.3	92.4
		MCWB	76.7	76.0	77.4	79.0	80.2	79.0	78.8	78.8	79.0	78.6	79.1	77.9
	2%	DB	91.9	94.4	96.2	95.2	93.1	92.1	91.0	90.9	90.5	90.4	90.9	91.0
		MCWB	76.5	76.2	77.3	79.4	79.8	79.1	78.7	78.6	78.7	78.7	78.6	77.6
	5%	DB	90.7	93.1	94.8	93.6	91.8	90.9	89.9	89.8	89.3	89.1	89.7	89.8
		MCWB	76.3	76.4	77.5	79.5	79.6	79.1	78.8	78.5	78.6	78.6	78.5	77.4
	10%	DB	89.4	91.6	93.0	92.0	90.5	89.6	88.8	88.6	88.0	87.6	88.2	88.5
		MCWB	76.1	76.2	77.5	79.5	79.5	79.2	78.7	78.5	78.5	78.4	78.3	77.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	79.9	81.0	82.5	81.9	81.6	80.9	80.8	80.8	80.7	80.7	80.2
		MCDB	88.1	90.8	92.0	92.5	90.7	89.5	88.3	88.6	88.3	88.2	88.7	88.6
	2%	WB	78.7	78.8	80.1	81.3	81.1	80.8	80.1	80.0	80.0	79.9	79.9	79.2
		MCDB	87.7	89.6	90.8	91.0	89.7	88.5	87.5	87.6	87.3	87.1	87.8	87.8
	5%	WB	78.0	78.0	79.3	80.7	80.6	80.2	79.8	79.5	79.3	79.2	79.2	78.5
		MCDB	87.0	88.7	89.9	90.1	89.0	87.6	87.1	86.9	86.4	86.3	87.0	87.0
	10%	WB	77.2	77.2	78.7	80.1	80.0	79.7	79.2	79.0	78.9	78.8	78.7	77.8
		MCDB	86.2	87.9	89.0	89.0	87.8	87.0	86.4	86.1	85.8	85.7	86.3	85.7
Mean Daily Temperature Range	5% DB	MDBR	13.5	16.5	17.2	15.2	13.1	12.7	12.4	12.0	11.6	11.7	12.1	12.0
		MCDBR	15.4	18.4	19.8	17.2	15.0	14.4	14.1	13.9	13.8	13.8	14.3	14.4
		MCWBR	5.1	5.4	5.6	5.1	4.6	4.4	4.6	4.3	4.3	4.5	4.7	4.8
	5% WB	MCDBR	13.7	16.5	17.2	15.4	13.6	13.1	13.0	12.8	12.6	12.5	13.1	12.9
		MCWBR	5.1	5.7	5.7	5.2	4.8	4.8	4.9	4.6	4.6	4.6	4.9	5.0
Clear-Sky Solar Irradiance	taub		0.453	0.466	0.476	0.473	0.455	0.463	0.467	0.468	0.458	0.454	0.450	0.454
	taud		2.323	2.257	2.257	2.275	2.351	2.349	2.325	2.310	2.335	2.349	2.357	2.336
	Ebn at Noon		274	274	271	267	266	260	261	265	272	274	273	272
	Edh at Noon		41	45	46	44	40	39	41	42	42	41	40	40
All-Sky Solar Radiation	RadAvg		1613	1890	1888	1831	1619	1564	1533	1561	1515	1475	1406	1383
	RadStd		114	110	161	90	93	79	72	60	60	106	100	127

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	+0.40	+0.25	N/A	+0.58	+0.40	+0.34	N/A	N/A	+169	+169	

Nomenclature: See separate page